

Kuan-Hung Chen

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Data Science Engineer specializing in Operational Excellence, automated VBA/Python pipelines, and predictive modeling.

Education

University of Warwick

MSc in Statistics (with Distinction)

Focus: *Statistical Inference, Data Analysis, Monte Carlo Methods, Machine Learning, Neural Networks (CNN, RNN, LSTM)*

Coventry, UK

Sep 2024 – Sep 2025

National Taiwan University

B.A. in Economics & B.A. in Foreign Languages and Literatures

Double Major, Overall GPA: 3.9 / 4.3

Focus: *Econometrics, Big Data Marketing, Translation*

Taipei, Taiwan

Sep 2019 – Jun 2024

Experience

Market Research Intern

KANTAR Worldpanel (now Worldpanel by Numerator)

Taipei, Taiwan

Jun 2023 – Apr 2024

- Engineered 10+ scalable Python and VBA data pipelines to act as an automation enabler, replacing manual Excel summaries and reporting workflows to drastically reduce operational waste.
- Analyzed complex consumer datasets to extract actionable insights, delivering reports and presentations that provided strategic input to management.
- Managed the end-to-end business solution lifecycle for a primary client account, facilitating cross-functional communication, creating training presentations, and delivering data-driven strategic insights.

Projects

MSc Dissertation (University of Warwick)

Probabilistic Weather Forecasting with Recursive Algorithms and Vine Copulas

Sep 2025

- Engineered an efficient vine copula-based deep learning algorithm utilizing PyTorch, demonstrating advanced quantitative modeling applicable to operational forecasting.
- Implemented a Quasi-Bayesian recursive density estimation algorithm to enable "online" learning and rapid metric monitoring.
- Designed the system to dynamically update predictions upon new data ingestion, streamlining data-driven decision-making in high-variance environments.

Open-Source Developer | **obsidian-2-jekyll**

Obsidian-to-Jekyll Static Content Pipeline

2026

- Engineered a Python pipeline converting Obsidian Markdown into Jekyll-compatible output, implementing incremental builds and asset syncing to ensure posts are up-to-date.
- Hardened the converter against real-world input with a 129-test suite spanning unit, integration, and subprocess-driven CLI test.
- Packaged the tool as a distributable composite GitHub Action exposing 12 inputs and 4 workflow outputs, publishing to a separate site repository through reviewable pull requests.

Machine Learning Project | **NLP Product Categorizer**

NLP Product Categorizer for zh-TW E-Commerce Context

- Engineered an NLP categorization framework utilizing TF-IDF vectorization and SVC/MNB classifiers, specifically optimized for the complex naming conventions of the zh-TW e-commerce market (mixing English, numeric, and Chinese characters).
- Developed and containerized a production-ready backend API using FastAPI and Docker, enabling seamless deployment and real-time model inference.
- Achieved 70% accuracy and F1 scores against highly noisy, error-prone real-world datasets, establishing a robust performance baseline and demonstrating strong practical viability.

Skills

Programming: SQL, Python, C++, R, VBA, JavaScript, Ruby

Machine Learning: Numpy, Pandas, Scipy, scikit-learn, CNN, RNN, LSTM, NLP, PyTorch

Data Analysis: Stata, Tableau, Time Series Forecasting, ETL pipeline

Web Development: Ruby on Rails, React, FastAPI, HTML, CSS

Tools: Bash, Docker, Git, $\text{\LaTeX}$, Excel, PowerPoint

Languages: Mandarin (Native), English (Professional; IELTS Academic 8.0), German (Intermediate)